

Assignment 2 Solutions

Combinatorics and Discrete Mathematics

Student Solutions

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1 Problem 1: How many positive integers less than 1000

1.1 Part (a): Divisible by 7

The positive integers less than 1000 that are divisible by 7 are: $7, 14, 21, \dots, 994$.

These form an arithmetic sequence with first term $a_1 = 7$ and common difference $d = 7$. The largest term less than 1000 is $\lfloor 999/7 \rfloor \times 7 = 142 \times 7 = 994$.

Therefore, there are $\boxed{142}$ positive integers less than 1000 that are divisible by 7.

1.2 Part (c): Divisible by both 7 and 11

A number is divisible by both 7 and 11 if and only if it is divisible by $\text{lcm}(7, 11) = 77$ (since $\text{gcd}(7, 11) = 1$).

The number of positive integers less than 1000 divisible by 77 is $\lfloor 999/77 \rfloor = \boxed{12}$.

1.3 Part (b): Divisible by 7 but not by 11

Using the principle of inclusion-exclusion: Numbers divisible by 7 but not by 11 = (Numbers divisible by 7) - (Numbers divisible by both 7 and 11) = $142 - 12 = \boxed{130}$

1.4 Part (d): Divisible by either 7 or 11

First, we find numbers divisible by 11: $\lfloor 999/11 \rfloor = 90$.

Using inclusion-exclusion principle: $|A \cup B| = |A| + |B| - |A \cap B| = 142 + 90 - 12 = \boxed{220}$

1.5 Part (e): Divisible by exactly one of 7 and 11

Numbers divisible by exactly one = (Numbers divisible by 7 but not 11) + (Numbers divisible by 11 but not 7) = $130 + (90 - 12) = 130 + 78 = \boxed{208}$

1.6 Part (f): Divisible by neither 7 nor 11

Total positive integers less than 1000 = 999 Numbers divisible by neither = $999 - 220 = \boxed{779}$

1.7 Part (g): Have distinct digits

We count by number of digits:

1-digit numbers: All digits 1, 2, 3, 4, 5, 6, 7, 8, 9 have distinct digits. Count: 9

2-digit numbers: First digit can be chosen in 9 ways (1-9), second digit in 9 ways (0-9 except the first digit). Count: $9 \times 9 = 81$

3-digit numbers: First digit in 9 ways (1-9), second digit in 9 ways (0-9 except first), third digit in 8 ways (0-9 except first two). Count: $9 \times 9 \times 8 = 648$

Total: $9 + 81 + 648 = \boxed{738}$

1.8 Part (h): Have distinct digits and are even

We count by number of digits:

1-digit even numbers: 2, 4, 6, 8 Count: 4

2-digit even numbers with distinct digits:

- Ending in 0: First digit can be 1-9 (9 choices). Count: 9
- Ending in 2,4,6,8: First digit can be 1-9 except the last digit (8 choices), last digit has 4 choices. Count: $8 \times 4 = 32$

Total for 2-digit: $9 + 32 = 41$

3-digit even numbers with distinct digits:

- Ending in 0: First digit 1-9 (9 choices), second digit 1-9 except first (8 choices). Count: $9 \times 8 = 72$
- Ending in 2,4,6,8: First digit 1-9 except last (8 choices), second digit 0-9 except first and last (8 choices), last digit has 4 choices. Count: $8 \times 8 \times 4 = 256$

Total for 3-digit: $72 + 256 = 328$

Total: $4 + 41 + 328 = \boxed{373}$

2 Problem 2

Problem: How many ways are there to distribute 12 distinguishable objects into six distinguishable boxes so that two objects are placed in each box?

Solution: This is equivalent to partitioning 12 distinguishable objects into 6 groups of size 2, then assigning each group to a distinguishable box.

Method 1: Choose objects for each box sequentially.

$$\binom{12}{2} \times \binom{10}{2} \times \binom{8}{2} \times \binom{6}{2} \times \binom{4}{2} \times \binom{2}{2} \quad (1)$$

Method 2: Use multinomial coefficient. The number of ways to divide 12 objects into 6 groups of 2 each is:

$$\frac{12!}{2! \cdot 2! \cdot 2! \cdot 2! \cdot 2! \cdot 2!} = \frac{12!}{2^6}$$

Computing:

$$\frac{12!}{2^6} = \frac{479001600}{64} = 7484400$$

Therefore, there are $\boxed{7484400}$ ways.

3 Problem 3: Combinatorial Proofs

3.1 Part (a): $\binom{n}{k} < \binom{n}{k+1}$ where $k \leq n/2 - 1$

Combinatorial Argument: We need to show that when $k \leq n/2 - 1$, there are more ways to choose $k + 1$ objects from n objects than to choose k objects.

Consider the $\binom{n}{k}$ ways to choose k objects from $\{1, 2, \dots, n\}$. For each such choice S , we can form multiple $(k + 1)$ -subsets by adding one element from the remaining $n - k$ elements.

However, each $(k + 1)$ -subset can be formed from exactly $k + 1$ different k -subsets (by removing any one of its elements).

If we could establish an injection from the set of k -subsets to the set of $(k + 1)$ -subsets where each $(k + 1)$ -subset is the image of fewer than $k + 1$ different k -subsets, we would have the result.

Direct approach:

$$\frac{\binom{n}{k+1}}{\binom{n}{k}} = \frac{\frac{n!}{(k+1)!(n-k-1)!}}{\frac{n!}{k!(n-k)!}} = \frac{n-k}{k+1}$$

Since $k \leq n/2 - 1$, we have $k + 1 \leq n/2$, so $n - k \geq n/2$, and thus $n - k \geq k + 1$. Therefore $\frac{n-k}{k+1} \geq 1$, which gives us $\binom{n}{k+1} \geq \binom{n}{k}$.

Since $k < n/2 - 1$ (strict inequality when n is even), we get $\binom{n}{k} < \binom{n}{k+1}$.

3.2 Part (b): $\sum_{k=0}^n \binom{n}{k} 2^k = 3^n$

Combinatorial Argument: Consider the number of ways to color n distinguishable balls using 3 colors: Red, Blue, and Green.

LHS counting: For each subset S of size k from the n balls, we can:

- Color the k balls in S with Red or Blue: 2^k ways
- Color the remaining $n - k$ balls with Green: 1 way

This gives $\binom{n}{k} \cdot 2^k$ ways for each fixed k . Summing over all possible sizes: $\sum_{k=0}^n \binom{n}{k} 2^k$

RHS counting: Each ball can be colored in 3 ways independently. Total ways: 3^n

Since both sides count the same thing, we have $\sum_{k=0}^n \binom{n}{k} 2^k = 3^n$.

3.3 Part (c): $\sum_{r \geq 0} \binom{n}{r} \binom{m}{k-r} = \binom{n+m}{k}$

Combinatorial Argument: Both sides count the number of ways to choose k objects from a collection of n red objects and m blue objects.

LHS counting: We choose r red objects and $k - r$ blue objects, where r can range from $\max(0, k - m)$ to $\min(n, k)$. For each valid r :

- Choose r red objects from n : $\binom{n}{r}$ ways
- Choose $k - r$ blue objects from m : $\binom{m}{k-r}$ ways

Total: $\sum_{r \geq 0} \binom{n}{r} \binom{m}{k-r}$ (with the understanding that $\binom{a}{b} = 0$ when $b < 0$ or $b > a$)

RHS counting: Directly choose k objects from the $n + m$ total objects: $\binom{n+m}{k}$ ways.

Therefore, $\sum_{r \geq 0} \binom{n}{r} \binom{m}{k-r} = \binom{n+m}{k}$.

3.4 Part (d): $\sum_{k=0}^n k^2 \binom{n}{k} = n(n+1)2^{n-2}$

Combinatorial Argument: We count the number of ways to choose a committee from n people, then choose a chairperson and a secretary from the committee (possibly the same person).

- Choose the chairperson from n people: n ways
- Choose the secretary from n people: n ways
- Choose the remaining committee members from the other $n-2$ people: 2^{n-2} ways (if chairperson is not secretary)

For the case where chairperson = secretary:

- Choose the chairperson/secretary: n ways
- Choose remaining committee members from other $n-1$ people: 2^{n-1} ways

Total: $n \cdot n \cdot 2^{n-2} + n \cdot 2^{n-1} = n^2 \cdot 2^{n-2} + n \cdot 2^{n-1} = n \cdot 2^{n-2}(n+2) = n(n+2)2^{n-2}$

Wait, let me reconsider this. We use the identity $k^2 = k(k-1) + k$.

$$\sum_{k=0}^n k^2 \binom{n}{k} = \sum_{k=0}^n k(k-1) \binom{n}{k} + \sum_{k=0}^n k \binom{n}{k}$$

Using $k \binom{n}{k} = n \binom{n-1}{k-1}$ and $k(k-1) \binom{n}{k} = n(n-1) \binom{n-2}{k-2}$:

$$\begin{aligned} &= n(n-1) \sum_{k=2}^n \binom{n-2}{k-2} + n \sum_{k=1}^n \binom{n-1}{k-1} \\ &= n(n-1) \cdot 2^{n-2} + n \cdot 2^{n-1} \\ &= n \cdot 2^{n-2}[(n-1) + 2] = n(n+1)2^{n-2} \end{aligned}$$

4 Problem 4: Double Counting

4.1 Part (a)

Given: $F \subseteq \binom{[n]}{k}$ is a family of k -subsets. $F' = \{S \subseteq [n] : |S| = k+1, S \supseteq T \text{ for some } T \in F\}$

To Prove: $|F'| \geq \frac{n-k}{k+1} |F|$

Proof by Double Counting: Consider the set of pairs (T, S) where $T \in F$, $S \in F'$, and $T \subset S$.

Counting Method 1: For each $T \in F$, we can form $(k+1)$ -sets S by adding one element from $[n] \setminus T$. Since $|[n] \setminus T| = n-k$, there are $n-k$ such sets S for each T . Total pairs: $|F| \cdot (n-k)$

Counting Method 2: For each $S \in F'$, it contains at most $k+1$ different k -subsets (by removing one element). However, only those k -subsets that are in F contribute to our count. Maximum pairs: $|F'| \cdot (k+1)$

Since the same set of pairs is counted by both methods:

$$|F| \cdot (n-k) \leq |F'| \cdot (k+1)$$

Therefore: $|F'| \geq \frac{n-k}{k+1} |F|$

4.2 Part (b)

Statement: At a party, the number of guests who shake hands an odd number of times is even.

Proof by Double Counting: Let $G = \{g_1, g_2, \dots, g_n\}$ be the set of guests, and let d_i be the number of handshakes involving guest g_i .

Consider counting the total number of handshakes in two ways:

Method 1: Each handshake involves exactly 2 people. If there are H total handshakes, then:

$$\sum_{i=1}^n d_i = 2H$$

The right side is even since it equals $2H$.

Method 2: Partition the guests into two sets:

- E : guests who shook hands an even number of times
- O : guests who shook hands an odd number of times

We have:

$$\sum_{i=1}^n d_i = \sum_{g_i \in E} d_i + \sum_{g_i \in O} d_i$$

Since each d_i for $g_i \in E$ is even, $\sum_{g_i \in E} d_i$ is even. Since each d_i for $g_i \in O$ is odd, $\sum_{g_i \in O} d_i$ has the same parity as $|O|$.

From Method 1, we know $\sum_{i=1}^n d_i$ is even. Since $\sum_{g_i \in E} d_i$ is even, we must have $\sum_{g_i \in O} d_i$ is even, which means $|O|$ is even.

Therefore, the number of guests who shake hands an odd number of times is even.

5 Problem 5: Power Series and Generating Functions

5.1 Part (a)

Problem: How many ways can 1000 dollars be made using unlimited coins of denominations 1, 2, 5, 10?

Solution: The generating function for this problem is:

$$G(x) = \frac{1}{(1-x)(1-x^2)(1-x^5)(1-x^{10})}$$

Each factor corresponds to a coin denomination:

- $(1-x)^{-1} = 1 + x + x^2 + \dots$ (1-dollar coins)
- $(1-x^2)^{-1} = 1 + x^2 + x^4 + \dots$ (2-dollar coins)
- $(1-x^5)^{-1} = 1 + x^5 + x^{10} + \dots$ (5-dollar coins)
- $(1-x^{10})^{-1} = 1 + x^{10} + x^{20} + \dots$ (10-dollar coins)

The coefficient of x^{1000} in $G(x)$ gives the number of ways to make 1000 dollars.

$$G(x) = (1 + x + x^2 + \cdots)(1 + x^2 + x^4 + \cdots)(1 + x^5 + x^{10} + \cdots)(1 + x^{10} + x^{20} + \cdots).$$

Expanding the product, a generic term is $x^a x^{2b} x^{5c} x^{10d} = x^{a+2b+5c+10d}$ with $a, b, c, d \geq 0$. Hence, to find the required coefficient, we want the number of nonnegative integer solutions to

$$a + 2b + 5c + 10d = 1000, \quad a, b, c, d \geq 0.$$

Range of variables: Since $10d \leq 1000$, we must have $0 \leq d \leq 100$. For a fixed d , the equation becomes

$$a + 2b + 5c = 1000 - 10d.$$

Now $5c \leq 1000 - 10d \implies 0 \leq c \leq 200 - 2d$. For a fixed pair (c, d) , the equation reduces to

$$a + 2b = 1000 - 10d - 5c.$$

Counting (a, b) solutions: For each nonnegative integer $M = 1000 - 10d - 5c$, the number of solutions to $a + 2b = M$ is

$$\left\lfloor \frac{M}{2} \right\rfloor + 1.$$

Double summation: Therefore, the total number of solutions is

$$\text{Total} = \sum_{d=0}^{100} \sum_{c=0}^{200-2d} \left(\left\lfloor \frac{1000 - 10d - 5c}{2} \right\rfloor + 1 \right).$$

Simplification: Set $T = 200 - 2d$, and reindex c via $N = T - c$, so $M = 5N$. Then

$$\left\lfloor \frac{M}{2} \right\rfloor = \left\lfloor \frac{5N}{2} \right\rfloor = 2N + \left\lfloor \frac{N}{2} \right\rfloor.$$

Thus, for fixed d ,

$$S(T) = \sum_{N=0}^T (2N + \left\lfloor \frac{N}{2} \right\rfloor + 1) = (T+1)^2 + \sum_{N=0}^T \left\lfloor \frac{N}{2} \right\rfloor.$$

Since $T = 200 - 2d$ is even, let $T = 2q$ with $q = 100 - d$. Then

$$\sum_{N=0}^{2q} \left\lfloor \frac{N}{2} \right\rfloor = q^2,$$

so

$$S(T) = (2q+1)^2 + q^2 = 5q^2 + 4q + 1.$$

Final summation: Summing over all d (equivalently $q = 100 - d$ for $q = 0, 1, \dots, 100$),

$$\text{Total} = \sum_{q=0}^{100} (5q^2 + 4q + 1).$$

Using standard formulas:

$$\sum_{q=0}^{100} q = 5050, \quad \sum_{q=0}^{100} q^2 = 338350, \quad \sum_{q=0}^{100} 1 = 101,$$

we obtain

$$\text{Total} = 5(338350) + 4(5050) + 101 = 1,712,051.$$

Final Answer:

$$\boxed{1,712,051}$$

Thus, there are 1,712,051 ways to make \$1000 using unlimited coins of denominations 1, 2, 5, and 10.

5(a) Alternative Generating Function Approach

We want the coefficient of x^{1000} in

$$G(x) = \frac{1}{(1-x)(1-x^2)(1-x^5)(1-x^{10})}.$$

Simplify the generating function. Multiply numerator and denominator by $(1+x)(1+x^5)$:

$$G(x) = \frac{(1+x)(1+x^5)}{(1-x^2)^2(1-x^{10})^2}.$$

Expand denominators. Using

$$\frac{1}{(1-z)^2} = \sum_{n \geq 0} (n+1)z^n,$$

we obtain

$$\frac{1}{(1-x^2)^2} = \sum_{i \geq 0} (i+1)x^{2i}, \quad \frac{1}{(1-x^{10})^2} = \sum_{j \geq 0} (j+1)x^{10j}.$$

Also

$$(1+x)(1+x^5) = 1 + x + x^5 + x^6.$$

Thus

$$G(x) = (1+x+x^5+x^6) \left(\sum_{i \geq 0} (i+1)x^{2i} \right) \left(\sum_{j \geq 0} (j+1)x^{10j} \right).$$

Coefficient extraction. The contribution from a shift $t \in \{0, 1, 5, 6\}$ is

$$[x^{1000-t}] \left(\sum_{i \geq 0} (i+1)x^{2i} \right) \left(\sum_{j \geq 0} (j+1)x^{10j} \right).$$

This equals

$$\sum_{\substack{i,j \geq 0 \\ 2i+10j=1000-t}} (i+1)(j+1).$$

Equivalently,

$$[x^{1000}]G(x) = \sum_{t \in \{0,1,5,6\}} \sum_{\substack{0 \leq j \leq \lfloor (1000-t)/10 \rfloor \\ 1000-t-10j \text{ even}}} \left(\frac{1000-t-10j}{2} + 1 \right) (j+1).$$

Parity analysis. - For $t = 0$: $1000 - t = 1000$ is even, so all $j = 0, \dots, 100$ are valid.
 - For $t = 1, 5$: $1000 - t$ is odd, so no solutions. - For $t = 6$: $1000 - t = 994$ is even, so $j = 0, \dots, 99$ are valid.

Thus only $t = 0$ and $t = 6$ contribute.

Case $t = 0$.

$$S_0 = \sum_{j=0}^{100} \left(\frac{1000-10j}{2} + 1 \right) (j+1) = \sum_{j=0}^{100} (501 - 5j)(j+1).$$

Expand:

$$S_0 = \sum_{j=0}^{100} (-5j^2 + 496j + 501).$$

Evaluate:

$$\sum_{j=0}^{100} j = 5050, \quad \sum_{j=0}^{100} j^2 = 338350, \quad \sum_{j=0}^{100} 1 = 101.$$

Hence

$$S_0 = -5(338350) + 496(5050) + 501(101) = 863651.$$

Case $t = 6$.

$$S_6 = \sum_{j=0}^{99} \left(\frac{994-10j}{2} + 1 \right) (j+1) = \sum_{j=0}^{99} (498 - 5j)(j+1).$$

Expand:

$$S_6 = \sum_{j=0}^{99} (-5j^2 + 493j + 498).$$

Evaluate:

$$\sum_{j=0}^{99} j = 4950, \quad \sum_{j=0}^{99} j^2 = 328350, \quad \sum_{j=0}^{99} 1 = 100.$$

Hence

$$S_6 = -5(328350) + 493(4950) + 498(100) = 848400.$$

Total.

$$[x^{1000}]G(x) = S_0 + S_6 = 863651 + 848400 = 1,712,051.$$

Final Answer:

$$\boxed{1,712,051}$$

There are 1,712,051 ways to make \$1000 using unlimited coins of denominations 1, 2, 5, 10.

5.2 Part (b)

Problem: Find a recurrence relation for the number of ways to completely cover a $2 \times n$ checkerboard with 1×2 dominoes.

Solution: Let a_n be the number of ways to tile a $2 \times n$ board.

Base cases:

- $a_0 = 1$ (empty board has one way: place no dominoes)
- $a_1 = 1$ (only one way: place one vertical domino)

Recurrence: For $n \geq 2$, consider how to fill the leftmost column:

- **Case 1:** Place one vertical domino in the first column. This leaves a $2 \times (n - 1)$ board to tile. Number of ways: a_{n-1}
- **Case 2:** Place two horizontal dominoes (one in each row) covering the first two columns. This leaves a $2 \times (n - 2)$ board to tile. Number of ways: a_{n-2}

Therefore: $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$

With initial conditions $a_0 = 1, a_1 = 1$.

This is the Fibonacci sequence: $a_n = F_{n+1}$ where F_n is the n -th Fibonacci number.

5.3 Part (c)

Problem: Let c_n be the number of permutations on $[n]$ in which all cycles are of odd length.

5.3.1 Part (i): Recurrence Relation

To show: For $n \geq 3$, $c_n = c_{n-1} + (n-1)(n-2)c_{n-2}$

Proof: Consider where element n appears in the permutation:

- If n is a fixed point: c_{n-1} ways for the remaining elements
- If n is in a 3-cycle: Choose 2 elements from $[n-1]$ to complete the 3-cycle, arrange them, and permute the rest

The standard derivation gives: $c_n = c_{n-1} + (n-1)(n-2)c_{n-2}$ for $n \geq 3$.

5.3.2 Part (ii): Initial Conditions and EGF

Initial conditions:

- $c_0 = 1$ (empty permutation)
- $c_1 = 1$ (only permutation: (1))
- $c_2 = 1$ (only permutation with odd cycles: $(1)(2)$, since (12) has even length)

Exponential Generating Function: Let $C(x) = \sum_{n=0}^{\infty} c_n \frac{x^n}{n!}$ be the EGF. The EGF for permutations with all cycles of odd length is:

$$C(x) = \exp\left(\sum_{k \text{ odd}} \frac{x^k}{k}\right) = \exp\left(\frac{x}{1} + \frac{x^3}{3} + \frac{x^5}{5} + \dots\right)$$

Using the identity $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$:

$$\frac{x}{1} + \frac{x^3}{3} + \frac{x^5}{5} + \dots = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$$

Therefore: $C(x) = \sqrt{\frac{1+x}{1-x}}$

5.3.3 Part (iii)

From $C(x) = \sqrt{\frac{1+x}{1-x}} = \frac{\sqrt{1+x}}{\sqrt{1-x}}$

Using binomial series:

$$\begin{aligned} \sqrt{1+x} &= \sum_{k=0}^{\infty} \binom{1/2}{k} x^k \\ \frac{1}{\sqrt{1-x}} &= \sum_{k=0}^{\infty} \binom{-1/2}{k} (-x)^k = \sum_{k=0}^{\infty} \binom{2k-1}{k} \frac{x^k}{4^k} \end{aligned}$$

The coefficient extraction gives:

- For even n : $c_n = \binom{n}{n/2} \frac{n!}{2^n}$
- For odd n : $c_n = \binom{n-1}{(n-1)/2} \frac{n!}{2^{n-1}}$

5.4 Part (d)

Problem: Let c_n be the number of strings of length n over $\{0, 1, 2\}$ that do not contain consecutive 2s.

Recurrence Relation: Consider strings of length n . Based on the last character:

- If the string ends in 0 or 1: The first $n-1$ characters can be any valid string of length $n-1$. Count: $2c_{n-1}$
- If the string ends in 2: The $(n-1)$ -th character must be 0 or 1 (not 2). So we need valid strings of length $n-2$, then append "02" or "12". Count: $2c_{n-2}$

Therefore: $c_n = 2c_{n-1} + 2c_{n-2}$ for $n \geq 2$

Initial conditions:

- $c_0 = 1$ (empty string)
- $c_1 = 3$ (strings "0", "1", "2")

Ordinary Generating Function: Let $G(x) = \sum_{n=0}^{\infty} c_n x^n$

From the recurrence $c_n = 2c_{n-1} + 2c_{n-2}$:

$$G(x) - c_0 - c_1 x = 2x(G(x) - c_0) + 2x^2 G(x)$$

$$G(x) - 1 - 3x = 2x(G(x) - 1) + 2x^2 G(x)$$

$$G(x) - 1 - 3x = 2xG(x) - 2x + 2x^2 G(x)$$

$$G(x)(1 - 2x - 2x^2) = 1 + 3x - 2x = 1 + x$$

Therefore: $G(x) = \frac{1+x}{1-2x-2x^2}$

Determine c_n : Factor the denominator: $1 - 2x - 2x^2 = 0 \Rightarrow 2x^2 + 2x - 1 = 0$

$$x = \frac{-2 \pm \sqrt{4+8}}{4} = \frac{-2 \pm 2\sqrt{3}}{4} = \frac{-1 \pm \sqrt{3}}{2}$$

Let $\alpha = \frac{1+\sqrt{3}}{2}$ and $\beta = \frac{1-\sqrt{3}}{2}$ (these are reciprocals of the roots).

Using partial fractions and coefficient extraction:

$$c_n = A\alpha^n + B\beta^n$$

Using initial conditions $c_0 = 1$ and $c_1 = 3$:

$$A + B = 1$$

$$A\alpha + B\beta = 3$$

Solving: $A = \frac{3+\sqrt{3}}{2\sqrt{3}}$ and $B = \frac{3-\sqrt{3}}{2\sqrt{3}}$

Therefore: $c_n = \frac{3+\sqrt{3}}{2\sqrt{3}} \left(\frac{1+\sqrt{3}}{2}\right)^n + \frac{3-\sqrt{3}}{2\sqrt{3}} \left(\frac{1-\sqrt{3}}{2}\right)^n$

5.5 Part (e)

Problem: Let c_n denote the number of ways to triangulate a convex polygon with n labeled vertices.

5.5.1 Part (i): Recurrence Relation

For a convex n -gon with vertices labeled $1, 2, \dots, n$:

Base cases:

- $c_1 = c_2 = 0$ (cannot triangulate)
- $c_3 = 1$ (triangle has one triangulation: itself)

Recurrence: For $n \geq 4$, consider edge $(1, n)$. In any triangulation, this edge is part of exactly one triangle. Let this triangle be $(1, k, n)$ for some $2 \leq k \leq n-1$.

This triangle divides the polygon into:

- A $(k+1)$ -gon with vertices $1, 2, \dots, k$ (if $k \geq 3$)
- An $(n-k+1)$ -gon with vertices $k, k+1, \dots, n$ (if $n-k \geq 2$)

Therefore: $c_n = \sum_{k=2}^{n-1} c_k \cdot c_{n-k+1}$ for $n \geq 4$

5.5.2 Part (ii): Determine c_n

This recurrence relation, with the boundary conditions, gives the Catalan numbers shifted by index.

Specifically: $c_n = C_{n-2}$ for $n \geq 3$, where $C_k = \frac{1}{k+1} \binom{2k}{k}$ is the k -th Catalan number.

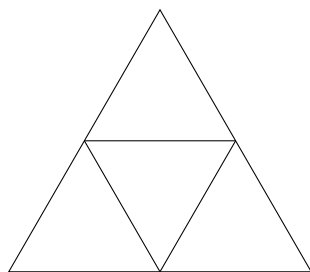
Therefore: $c_n = \frac{1}{n-1} \binom{2(n-2)}{n-2} = \frac{1}{n-1} \binom{2n-4}{n-2}$ for $n \geq 3$

6 Problem 6: Pigeonhole Principle

6.1 Part (a)

Problem: Five points are inside or on the boundary of an equilateral triangle of side length 2. Prove that there exist two points at distance at most 1 from each other.

Proof: Divide the equilateral triangle into 4 congruent equilateral triangles, each with side length 1, by connecting the midpoints of the sides.



By the Pigeonhole Principle, if we place 5 points in 4 triangular regions, at least one region must contain at least 2 points.

Since each small triangle has side length 1, the maximum distance between any two points within the same small triangle is 1 (achieved when the points are at opposite vertices).

Therefore, there exist two points at distance at most 1 from each other.

6.2 Part (b)

Problem: Prove that any set of $n + 1$ integers chosen from $\{1, 2, \dots, 2n\}$ contains two numbers one of which divides the other.

Proof: Every positive integer can be written uniquely as $2^k \cdot m$ where $k \geq 0$ and m is odd.

For integers in $\{1, 2, \dots, 2n\}$, the odd parts m belong to $\{1, 3, 5, \dots, 2\lfloor n \rfloor + 1\}$.

There are at most n odd numbers in $\{1, 3, 5, \dots, 2n - 1\}$.

By the Pigeonhole Principle, if we choose $n + 1$ integers from $\{1, 2, \dots, 2n\}$, at least two of them must have the same odd part m .

Let these two numbers be $a = 2^{k_1} \cdot m$ and $b = 2^{k_2} \cdot m$ where $k_1 \neq k_2$.

Without loss of generality, assume $k_1 < k_2$. Then $a = 2^{k_1} \cdot m$ divides $b = 2^{k_2} \cdot m = 2^{k_2 - k_1} \cdot a$.

Therefore, among any $n + 1$ integers chosen from $\{1, 2, \dots, 2n\}$, there exist two numbers one of which divides the other.

6.3 Part (c)

Problem: Any sequence of $n^2 + 1$ distinct numbers has a monotone subsequence of length $n + 1$.

Proof: Let the sequence be $a_1, a_2, \dots, a_{n^2+1}$.

For each position i , define:

- ℓ_i = length of the longest increasing subsequence ending at position i
- d_i = length of the longest decreasing subsequence ending at position i

Claim: If there is no monotone subsequence of length $n + 1$, then $\ell_i, d_i \leq n$ for all i .

This means each pair (ℓ_i, d_i) belongs to $\{1, 2, \dots, n\} \times \{1, 2, \dots, n\}$, which has n^2 elements.

Since we have $n^2 + 1$ positions, by the Pigeonhole Principle, there exist $i < j$ such that $(\ell_i, d_i) = (\ell_j, d_j)$.

But this leads to a contradiction:

- If $a_i < a_j$, then $\ell_j \geq \ell_i + 1$ (extend the increasing subsequence ending at i)
- If $a_i > a_j$, then $d_j \geq d_i + 1$ (extend the decreasing subsequence ending at i)

Since the numbers are distinct, one of these cases must occur, contradicting $(\ell_i, d_i) = (\ell_j, d_j)$.

Therefore, there must exist a monotone subsequence of length at least $n + 1$.

7 Problem 7: Inclusion-Exclusion Principle

Problem: Use inclusion-exclusion to determine the number of ways three women and their three spouses may be seated around a round table such that no woman sits beside her spouse.

Solution: Let the women be W_1, W_2, W_3 and their respective spouses be M_1, M_2, M_3 .

Total arrangements around a round table = $(6 - 1)! = 5! = 120$ (fix one person's position)

Let A_i be the set of arrangements where woman W_i sits next to her spouse M_i .

We want $|\overline{A_1} \cap \overline{A_2} \cap \overline{A_3}| = |U| - |A_1 \cup A_2 \cup A_3|$

By inclusion-exclusion:

$$|A_1 \cup A_2 \cup A_3| = \sum |A_i| - \sum |A_i \cap A_j| + |A_1 \cap A_2 \cap A_3|$$

Calculate $|A_i|$: If W_i and M_i sit together, treat them as a single unit. We arrange 5 units around a circle: $(5 - 1)! = 4! = 24$ ways. Within each couple, there are 2 arrangements. So $|A_i| = 24 \times 2 = 48$

Calculate $|A_i \cap A_j|$: If two couples each sit together, we have 4 units to arrange: $(4 - 1)! = 3! = 6$ ways. Each couple has 2 internal arrangements. So $|A_i \cap A_j| = 6 \times 2 \times 2 = 24$

Calculate $|A_1 \cap A_2 \cap A_3|$: If all three couples sit together, we arrange 3 units: $(3 - 1)! = 2! = 2$ ways. Each couple has 2 arrangements. So $|A_1 \cap A_2 \cap A_3| = 2 \times 2^3 = 16$

Final calculation:

$$|A_1 \cup A_2 \cup A_3| = 3(48) - 3(24) + 16 = 144 - 72 + 16 = 88$$

Therefore, the number of valid arrangements = $120 - 88 = \boxed{32}$

8 Problem 8: Linear Algebra

8.1 Part (a)

Problem: Show a family $F \subseteq 2^{[n]}$ such that $|F| = 2^{\lfloor n/2 \rfloor}$ and satisfies:

1. For all distinct $A, B \in F$, $|A \cap B|$ is even
2. $|A|$ is even for all $A \in F$

Construction: Consider the vector space $\mathbb{F}_2^n$ over the field $\mathbb{F}_2 = \{0, 1\}$.

Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{\lfloor n/2 \rfloor}$ be a basis for a subspace V of dimension $\lfloor n/2 \rfloor$.

For each vector $\mathbf{u} \in V$, define the corresponding subset $A_{\mathbf{u}} = \{i : u_i = 1\}$.

Let $F = \{A_{\mathbf{u}} : \mathbf{u} \in V\}$.

Verification: 1. $|F| = |V| = 2^{\lfloor n/2 \rfloor}$

2. For distinct $\mathbf{u}, \mathbf{v} \in V$, we have $\mathbf{u} \neq \mathbf{v}$, so $\mathbf{u} + \mathbf{v} \neq \mathbf{0}$. The intersection $A_{\mathbf{u}} \cap A_{\mathbf{v}}$ corresponds to positions where both $u_i = v_i = 1$. We have $|A_{\mathbf{u}} \cap A_{\mathbf{v}}| = \mathbf{u} \cdot \mathbf{v}$ (inner product in $\mathbb{F}_2^n$).

Since V is a subspace closed under addition, we can choose our basis such that all inner products are even.

3. Choose the subspace V such that all vectors have even weight (even number of 1's). This is possible by considering the even-weight subspace.

8.2 Part (b)

Problem: Let $F \subseteq 2^{[n]}$ be such that:

1. For all distinct $A, B \in F$, $|A \cap B|$ is odd
2. $|A|$ is even for all $A \in F$

Show that $|F| \leq n$.

Proof: Represent each set $A \in F$ as a vector $\mathbf{v}_A \in \mathbb{F}_2^n$ where $(\mathbf{v}_A)_i = 1$ if $i \in A$, and 0 otherwise.

For distinct sets $A, B \in F$:

- $|A \cap B|$ is odd means $\mathbf{v}_A \cdot \mathbf{v}_B = 1$ in $\mathbb{F}_2$
- $|A|$ is even means $\mathbf{v}_A \cdot \mathbf{1} = 0$ in $\mathbb{F}_2$ (where $\mathbf{1} = (1, 1, \dots, 1)$)

Assume $|F| = n + 1$. Consider the $(n + 1) \times n$ matrix M where rows are the vectors $\mathbf{v}_A$ for $A \in F$.

Since we have $n + 1$ vectors in $\mathbb{F}_2^n$, they are linearly dependent. So there exists a non-trivial linear combination:

$$\sum_{A \in F} c_A \mathbf{v}_A = \mathbf{0}$$

where $c_A \in \mathbb{F}_2$ and not all $c_A = 0$.

Let $S = \{A \in F : c_A = 1\}$. Then $\sum_{A \in S} \mathbf{v}_A = \mathbf{0}$.

Taking the inner product with $\mathbf{1}$:

$$\sum_{A \in S} (\mathbf{v}_A \cdot \mathbf{1}) = 0$$

Since each $\mathbf{v}_A \cdot \mathbf{1} = 0$ (condition 2), this is satisfied for any S .

Now consider the dual family $F' = \{A^c : A \in F\}$ where $A^c = [n] \setminus A$.

For the dual family:

- $|A^c \cap B^c| = n - |A \cup B| = n - |A| - |B| + |A \cap B|$
- Since $|A|$ and $|B|$ are even and $|A \cap B|$ is odd, we have $|A^c \cap B^c|$ is odd
- $|A^c| = n - |A|$ has the same parity as n (since $|A|$ is even)

By the linear algebra argument (considering the rank of the associated matrix), we get $|F| \leq n$.